

OMKAR KADAM

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OBJECTIVE

I recently completed my BE in Information Technology from Terna Engineering College (Mumbai University, 2026), and I work as a Full Stack Engineer at Riamona Luxury & Fashion Brands, where I build full-stack products end to end – writing the code, the tests, the CI/CD, and running them in production. Day to day I work with React, Node.js, Python, Prisma, and PostgreSQL. In 2026 I placed 7th of 31,000+ teams at the Meta PyTorch Hackathon, top 10 of 15,000+ at the ET AI Hackathon 2.0, and top 7 of 7,000+ at the iQOO Hackathon Pune City Battle, and co-authored a research paper now under review at *Discover Internet of Things* (Springer Nature). I'm looking for full-stack, platform, or ML engineering roles.

EDUCATION

Terna Engineering College, Nerul – BE Information Technology, Mumbai University	Nov 2022 – June 2026
D.A.V. Public School, Airoli – CBSE 12th Science	June 2020 – July 2022
D.A.V. Public School, Airoli – CBSE 10th	May 2020

EXPERIENCE

Full Stack Engineer – Riamona Luxury and Fashion Brands Pvt. Ltd., *Navi Mumbai* Jan 2026 – Present

- Build and ship full-stack features for several internal products using React, Node.js, Python, and PostgreSQL, along with the REST/GraphQL APIs and database schemas behind them
- Write unit, integration, and end-to-end tests for what I build, and maintain the GitHub Actions and Railway pipelines that run them on every push
- Handle deployments, monitoring, and production issues across staging and production, and work with the team to turn requirements into shipped features

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Kotlin, Java, SQL, HTML, CSS
AI Engineering	RAG, LLM APIs, MCP, LangChain, LangGraph, pgvector, OpenEnv, Gradio, Hugging Face (Transformers, TRL, PEFT, Spaces), LoRA and GRPO fine-tuning
ML & Data	PyTorch, TensorFlow, Scikit-learn, NLTK, OpenCV, MediaPipe, Pandas, NumPy, Streamlit, Deep Learning, NLP, Computer Vision, Reinforcement Learning
Web & Frameworks	React, Next.js, Node.js, Express.js, NestJS, Flask, FastAPI, Prisma, TailwindCSS, Vite, Three.js, WebSockets
Mobile	Android, Jetpack Compose, CameraX, Room, Firebase (Auth, Firestore)
Databases	PostgreSQL, PostGIS, MySQL, MongoDB, Supabase, Redis, SQLAlchemy
Cloud & DevOps	Google Cloud, AWS, Docker, Railway, Vercel, GitHub Actions, Git
APIs & Integration	REST APIs, GraphQL, JWT, OAuth, Shopify Admin API, BeautifulSoup
Blockchain	Solidity, Web3.js, Hardhat
Testing & Monitoring	pytest, Vitest, Playwright, Sentry

PROJECTS

- **VayuNetra – AI-Powered Urban Air Quality Intelligence (August 2026):** A LangGraph multi-agent platform that turns air quality readings into attributed, delivered interventions across 10 Indian cities (top 10 of 15,000+ teams, ET AI Hackathon 2.0): per-km² PM2.5 source attribution (GBM + SHAP), 72-hour calibrated forecasts, RAG-cited enforcement dossiers with draft notice PDFs, and citizen advisories in 8 languages over app, Telegram, and IVR – 16,529 modeled cells on zero-cost infrastructure. [GitHub](#)

- **Chakravyuh – Multi-Agent RL for UPI Fraud Detection (April 2026):** A five-agent OpenEnv RL environment (scammer, victim, analyzer, bank monitor, regulator) for UPI fraud (7th of 31,000+ teams, Meta PyTorch Hackathon). A Qwen2.5-7B analyzer fine-tuned with LoRA and GRPO reaches 99.3% detection at 6.7% false positives (36% before a reward-hacking fix), statistically tied with Llama-3.3-70B at 10× fewer parameters. Built the evaluation suite (frontier baselines, significance tests, leakage audit) and the Gradio demo. [GitHub](#)
- **Fitmon – Camera-Refereed Fitness Game (September 2026):** An Android fitness game where the front camera grades every rep (top 7 of 7,000+ teams, iQOO Pune City Battle). On-device MediaPipe pose tracking scores depth, range, tempo, and alignment, turned into damage against an adaptive boss; an on-device Gemma 3n coach cites only measured data. Kotlin, Jetpack Compose, CameraX, Room, Firebase; 836 JVM tests. [GitHub](#)
- **PyTorch Training Run Debugger (April 2026):** An OpenEnv RL environment (Meta PyTorch Hackathon, Round 1) where AI agents debug broken training runs: 7 real failures (exploding/vanishing gradients, data leakage, overfitting, BatchNorm in eval mode, training-loop bugs, bad LR schedules), with rewards gated on what the agent has inspected. FastAPI, WebSockets, Hugging Face Spaces; 245 tests, 95% coverage. [GitHub](#)
- **V2V Communication for Blind Spot Detection and Accident Prevention (May 2026):** A simulation where vehicles warn each other about blind spots and likely collisions before they happen. Built the messaging between vehicles, the sensor data processing, and a risk score that decides when to alert the driver. The work became a research paper, now under review at *Discover Internet of Things* (Springer Nature). [GitHub](#)
- **Smart PUC (May 2026):** A blockchain-based system for monitoring vehicle emissions and PUC compliance in India. OBD devices sign live emission telemetry, testing stations validate it, and the records go on-chain so no single party can fake the data. Tracks all five Bharat Stage VI pollutants with physics-based models and an ML ensemble for fraud detection, and issues emission certificates as NFTs. Built with Solidity smart contracts (Hardhat, OpenZeppelin), a FastAPI/Web3.py backend, and a web dashboard. [GitHub](#)

PUBLICATIONS

Vehicle-to-Vehicle Communication for Blind Spot Detection and Accident Prevention Using Severity-Gated Collision Risk Indexing – Co-author, manuscript submitted to *Discover Internet of Things* (Springer Nature); currently under peer review. April 2026

LEADERSHIP / EXTRA-CURRICULAR ACTIVITIES

- **7th Place – Meta PyTorch Hackathon (Scaler School of Technology, Bengaluru, Apr 2026):** Out of 31,000+ teams: online round with *PyTorch Training Run Debugger*, offline finals (top 800) with *Chakravyuh*.
- **Top 10 – ET AI Hackathon 2.0 (The Economic Times, Hyderabad, Aug 2026):** Placed in the top 10 of 15,000+ teams with *VayuNetra*, presented at the Hyderabad finale.
- **Top 7 – iQOO Hackathon 2026, Pune City Battle (iQOO, Pune, Sept 2026):** Placed in the top 7 of 7,000+ teams in the HealthTech track with *Fitmon*.
- **Finalist – Avishkar Research Project Competition 2025, Mumbai University:** Reached the finals of Mumbai University’s inter-collegiate research competition with the V2V research project.
- **Computer Society of India, Terna Chapter – Advisor (Oct 2025 – Present), Vice Chairperson (Oct 2024 – Oct 2025), Technical Executive (Nov 2023 – Oct 2024):** Rose to Vice Chairperson, leading a 40-member team that ran 10+ AI/ML workshops and seminars.
- **Training & Placement Cell, Terna – Deputy Secretary (Nov 2024 – Oct 2025), Hospitality Committee Member (Nov 2023 – Nov 2024):** Worked with 20+ recruiters and ran mock interviews to get students placement-ready.
- **Revive Cultural Fest, Terna – Hospitality HOD (Jan 2025 – Jan 2026), Senior Member (Jan 2024 – Jan 2025), Committee Member (Dec 2022 – Jan 2024):** Ran hospitality for 1,000+ attendees and guest judges with 50+ volunteers.